

ECON 108

INTRODUCTION TO ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

Extensions of demand and supply analysis

CHAPTER IV

CHAPTER OUTLINE

Changes in Demand and Supply

The Rationing Function of Prices

The Policy of Government-Imposed Price Controls

The Policy of Controlling Rents

Price Floors in Agriculture

Price Floors in the Labor Market

Quantity Restrictions

The Price System and Markets

Price system or market system

- An economic system in which relative prices are constantly changing to reflect changes in supply and demand:
- The prices are signals as to what is relatively scarce and relatively abundant.
- Prices provide information to individuals and businesses.

The Price System and Markets (cont'd)

Voluntary exchange

- An act of trading between individuals in the price system
- Makes both parties to the trade subjectively better off

The Price System and Markets (cont'd)

Transaction Costs

- All of the costs associated with exchange
- Including:
 - The informational costs of finding out the price and quality, service record, and durability of a product
 - The cost of contracting and enforcing that contract

The Price System and Markets (cont'd)

The role of middlemen

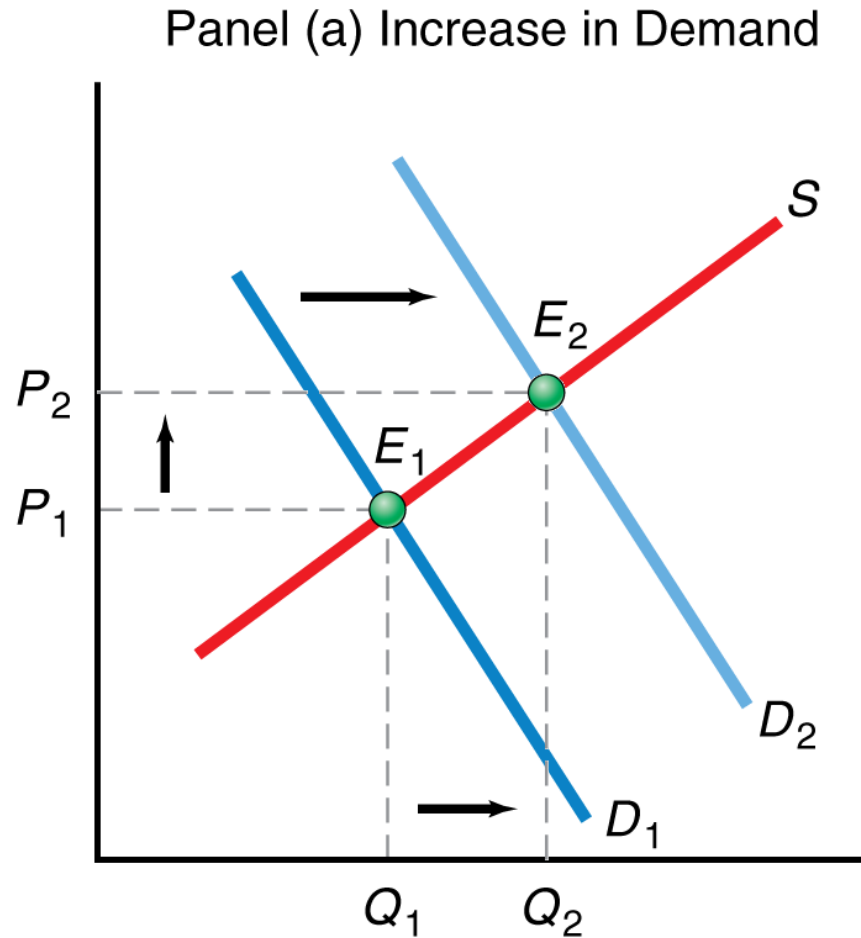
- Middlemen (intermediaries) or brokers reduce transaction costs by providing information to buyers and sellers
- Examples
 - Real estate brokers
 - Stock brokers
 - Consignment shops
 - Car dealerships

Changes in Demand and Supply

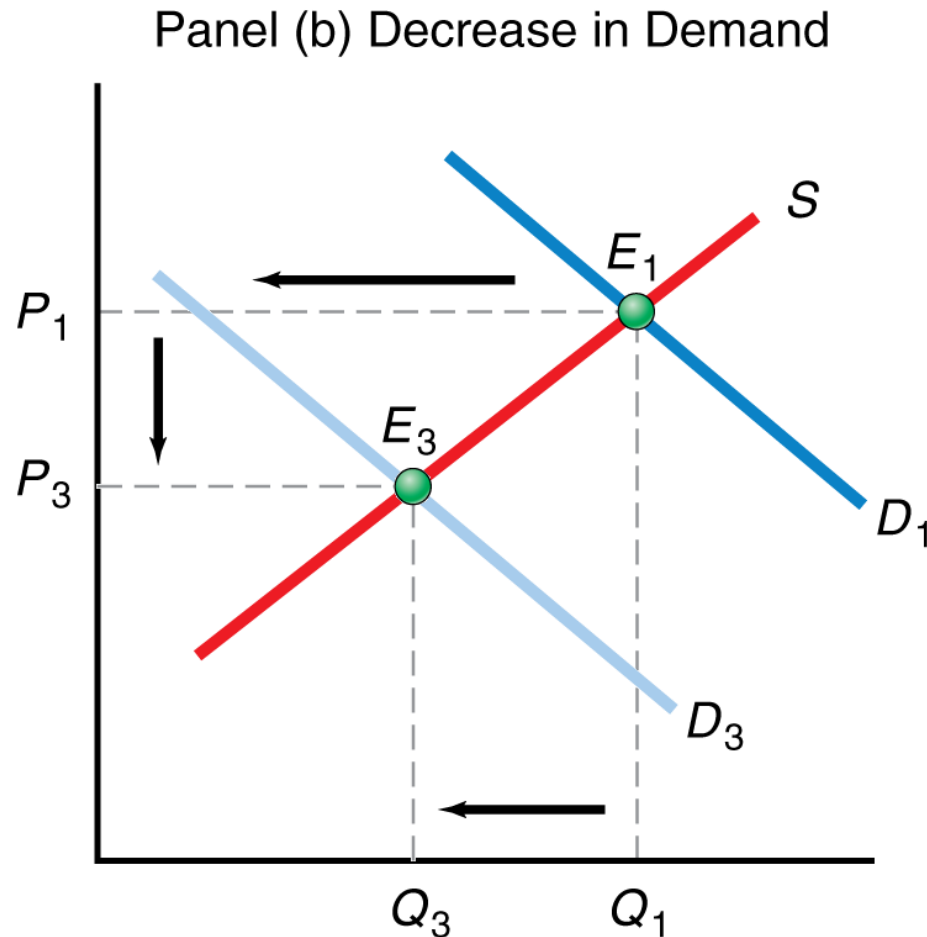
Changes in supply and demand create a disequilibrium.

The market price and quantity adjust to a new equilibrium.

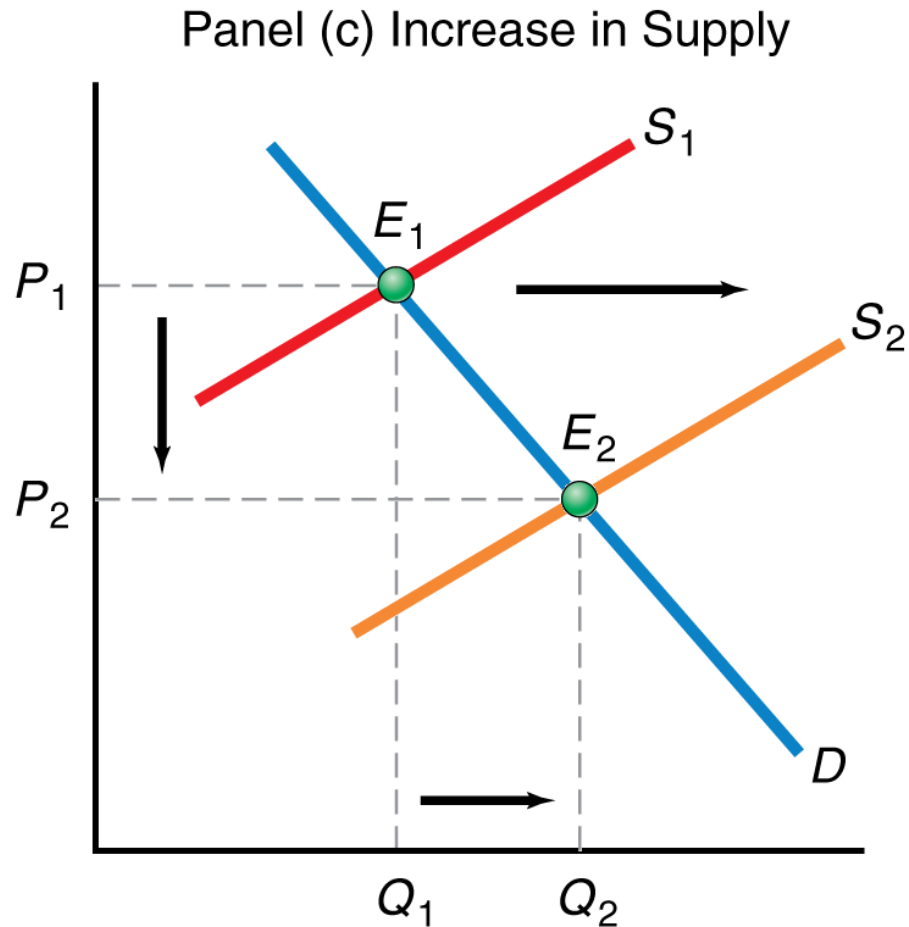
Shifts in Demand and in Supply: Determinate Results, Panel (a)



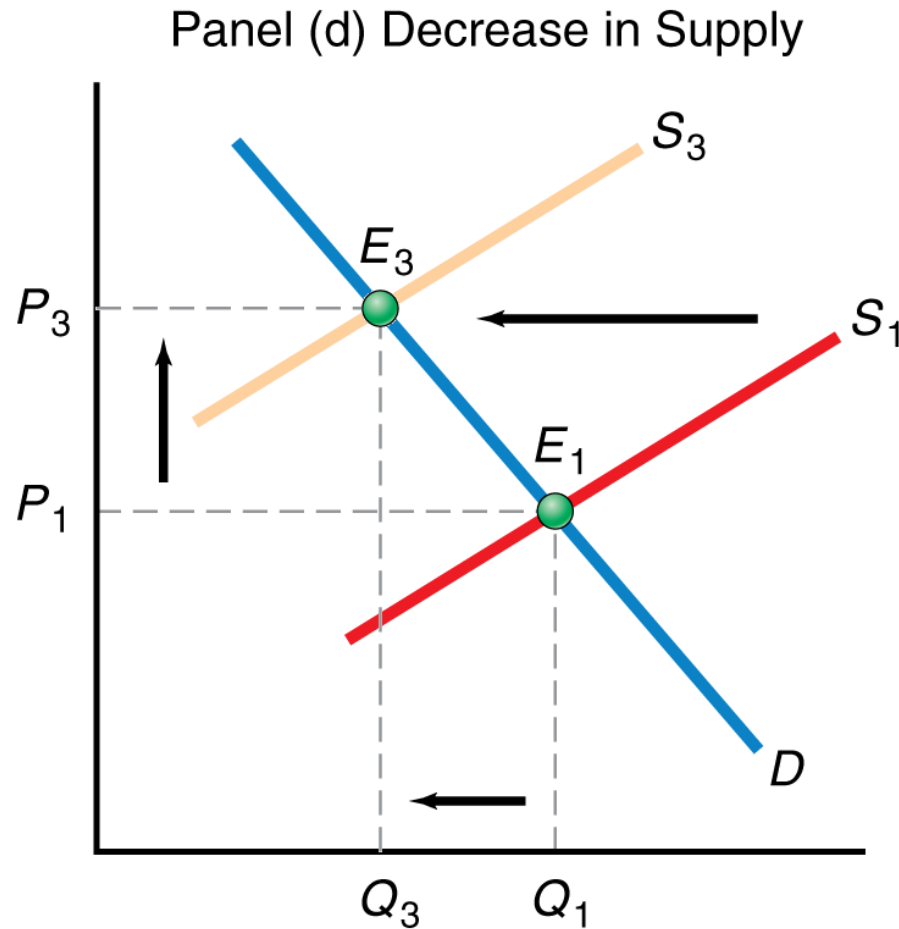
Shifts in Demand and in Supply: Determinate Results, Panel (b)



Shifts in Demand and in Supply: Determinate Results, Panel (c)



Shifts in Demand and in Supply: Determinate Results, Panel (d)



Changes in Demand and Supply (cont'd)

Summary

- Increases in *demand* increase equilibrium price and quantity
- Decreases in *demand* decrease equilibrium price and quantity

Changes in Demand and Supply (cont'd)

Summary

- Increases in *supply* decrease equilibrium price and increase equilibrium quantity
- Decreases in *supply* increase equilibrium price and decrease equilibrium quantity

Changes in Demand and Supply (cont'd)

When both demand and supply change

- If both the supply and demand curves shift simultaneously, the outcome is indeterminate for either equilibrium price or equilibrium quantity
- The resulting effect depends upon how much each curve shifts

Changes in Demand and Supply (cont'd)

When both demand & supply increase

- Change in equilibrium price is indeterminate
- Equilibrium quantity increases unambiguously

When both demand & supply decrease

- Change in equilibrium price is indeterminate
- Equilibrium quantity decreases unambiguously

Changes in Demand and Supply (cont'd)

When supply decreases & demand increases

- Equilibrium price increases
- The change in the equilibrium quantity is uncertain without more information

When supply increases & demand decreases

- Equilibrium price decreases
- The change in the equilibrium quantity is uncertain without more information

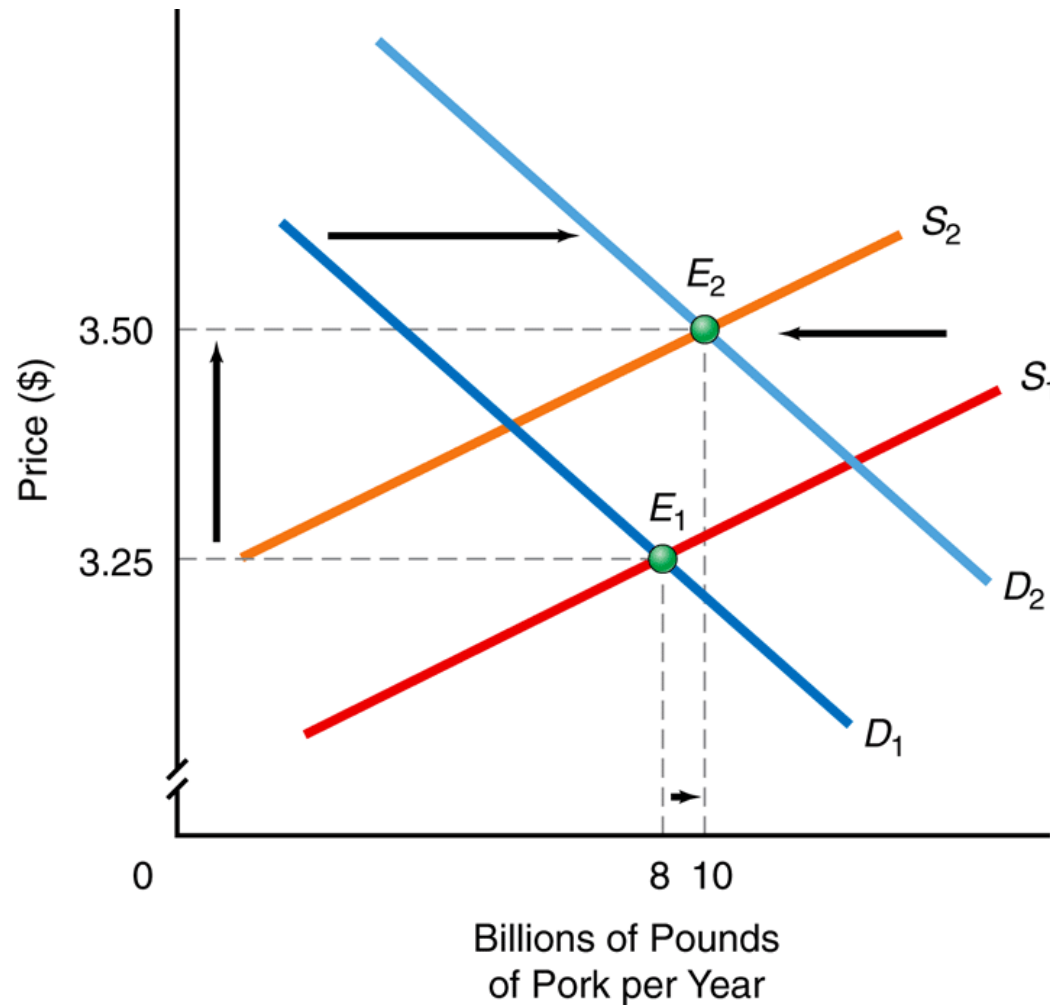
International Example: What Accounts for Rising Pork Prices in China?

Since the early 2000s, Chinese pork prices have surged.
Why?

Pork is a normal good, so that rising Chinese incomes had raised the demand for pork.

Meanwhile, rising prices of feed for hogs and higher prices of land to raise hogs contributed to a reduction in the supply of pork.

The Effects of a Simultaneous Decrease in Pork Supply and Increase in Pork Demand



Changes in Demand and Supply (cont'd)

Price Flexibility

Prices quite flexible in some markets can be less flexible in other market scenarios.

- May take the form of subtle adjustments such as hidden payments, quality changes
- May not reach equilibrium right away

Changes in Demand and Supply (cont'd)

Adjustment speed

- Market characteristics influence adjustment speed
- Markets may overshoot in the adjustment process
- Markets are subject to energy shocks, labor strikes, severe weather

The Rationing Function of Prices

Synchronization of decisions of buyers and sellers that leads to equilibrium is called the *rationing function of prices*

The Rationing Function of Prices (cont'd)

Methods of non-price rationing

- Rationing by queues (waiting in line)
- Rationing by random assignment or coupons

The Rationing Function of Prices (cont'd)

The essential role of rationing

- Implied by the presence of scarcity
- Price vs. non-price rationing mechanism:
 - Price rationing leads to the most efficient use of available resources
 - All gains from mutually beneficial trade are captured in a freely rationing price system

The Policy of Government-Imposed Price Controls

Price Controls

- Government-mandated minimum or maximum prices

Price Ceiling

- A legal maximum price

Price Floor

- A legal minimum price

The Policy of Government-Imposed Price Controls (cont'd)

Price ceiling and black markets

- A price ceiling may prevent the equilibrium price from being achieved if it is above the ceiling price
- A price ceiling that is set below the market clearing price creates a shortage

The Policy of Government-Imposed Price Controls (cont'd)

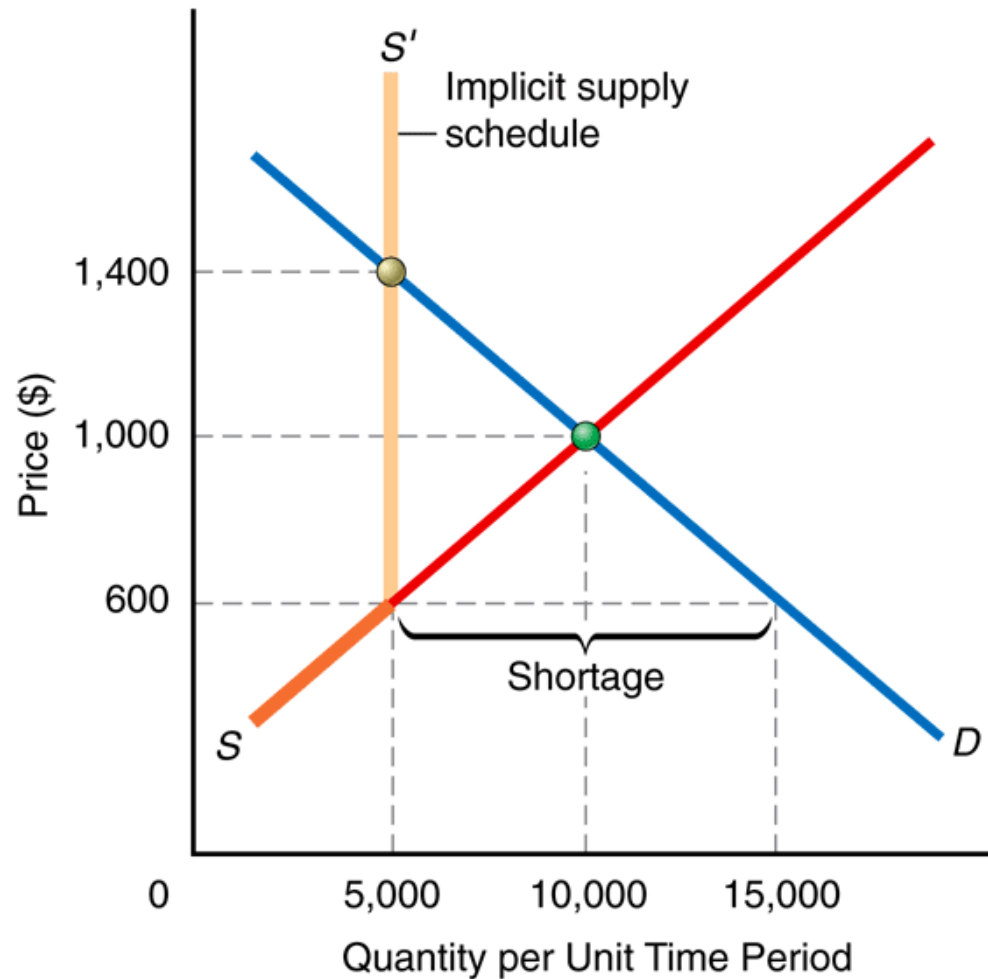
Non-Price Rationing Devices

- All methods used to ration scarce goods that are price-controlled

Black Market

- A market in which price-controlled goods are sold at an illegally high price

Black Markets for Portable Electric Generators



The Policy of Controlling Rents

The functions of rental prices

1. Promote the efficient maintenance and construction of housing
2. Allocate existing housing
3. Ration the use of housing

The Policy of Controlling Rents (cont'd)

Rent controls and construction

- Controls discourage construction
 - With a 16% vacancy rate and no controls, Dallas recently built 11,000 new rental units
 - With a 1.6% vacancy rate and controls, San Francisco recently built 2,000 new rental units

The Policy of Controlling Rents (cont'd)

Effects on the existing supply of housing and current use of housing

- Property owners cannot recover costs
 - Maintenance, repairs, capital improvements
- Rations the current use of housing
 - Reduces mobility, e.g., New York's "housing gridlock"

The Policy of Controlling Rents (cont'd)

Attempts to evade rent controls

- Forcing tenants to leave
- Tenants subletting apartments
- Housing courts

The Policy of Controlling Rents (cont'd)

Who gains and who loses from rent controls?

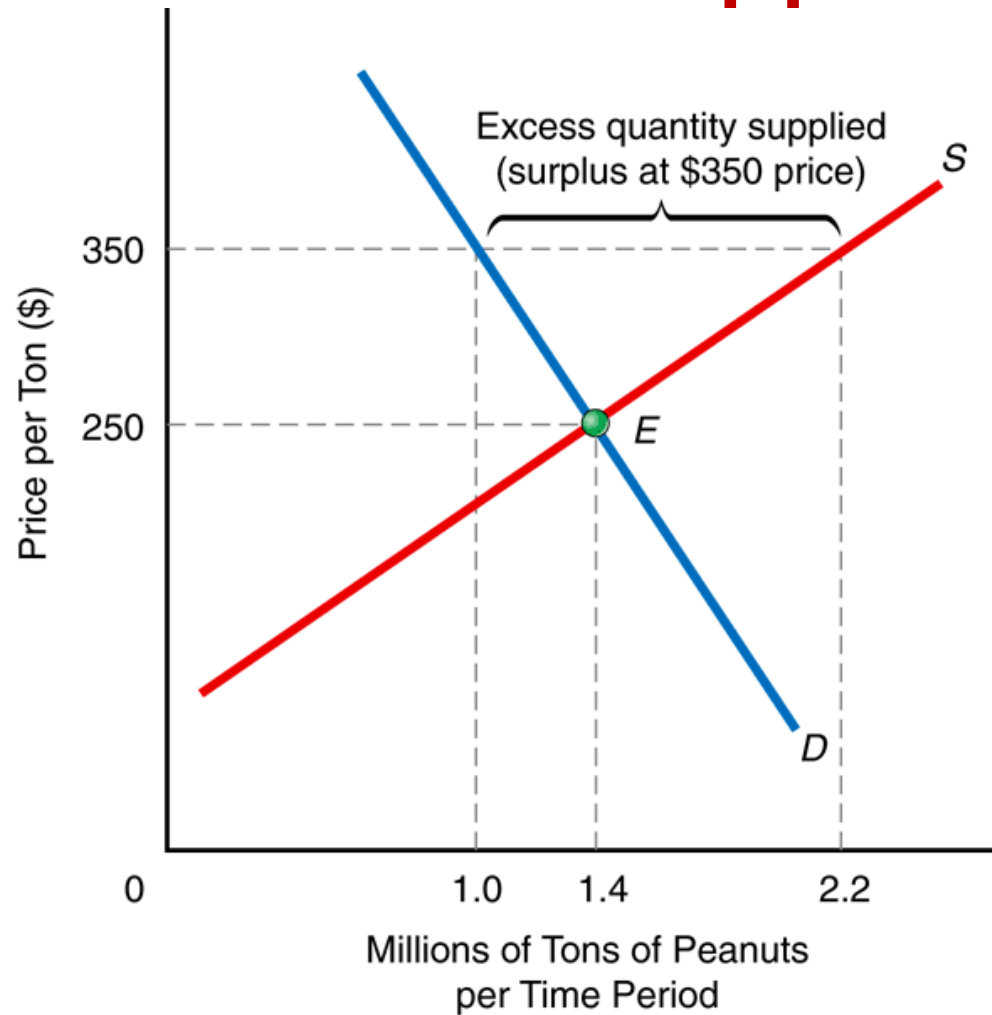
- Losers
 - Property owners
 - Low-income individuals
- Gainers
 - Upper-income professionals

Price Floors in Agriculture

Support Price

- The government chooses a price floor for a product and then acts to ensure that the price of the product never falls below the support level
- Associated with many agricultural products
- A price floor that is set above the market clearing price results in a surplus.

Agricultural Price Supports



Price Floors in Agriculture (cont'd)

Questions

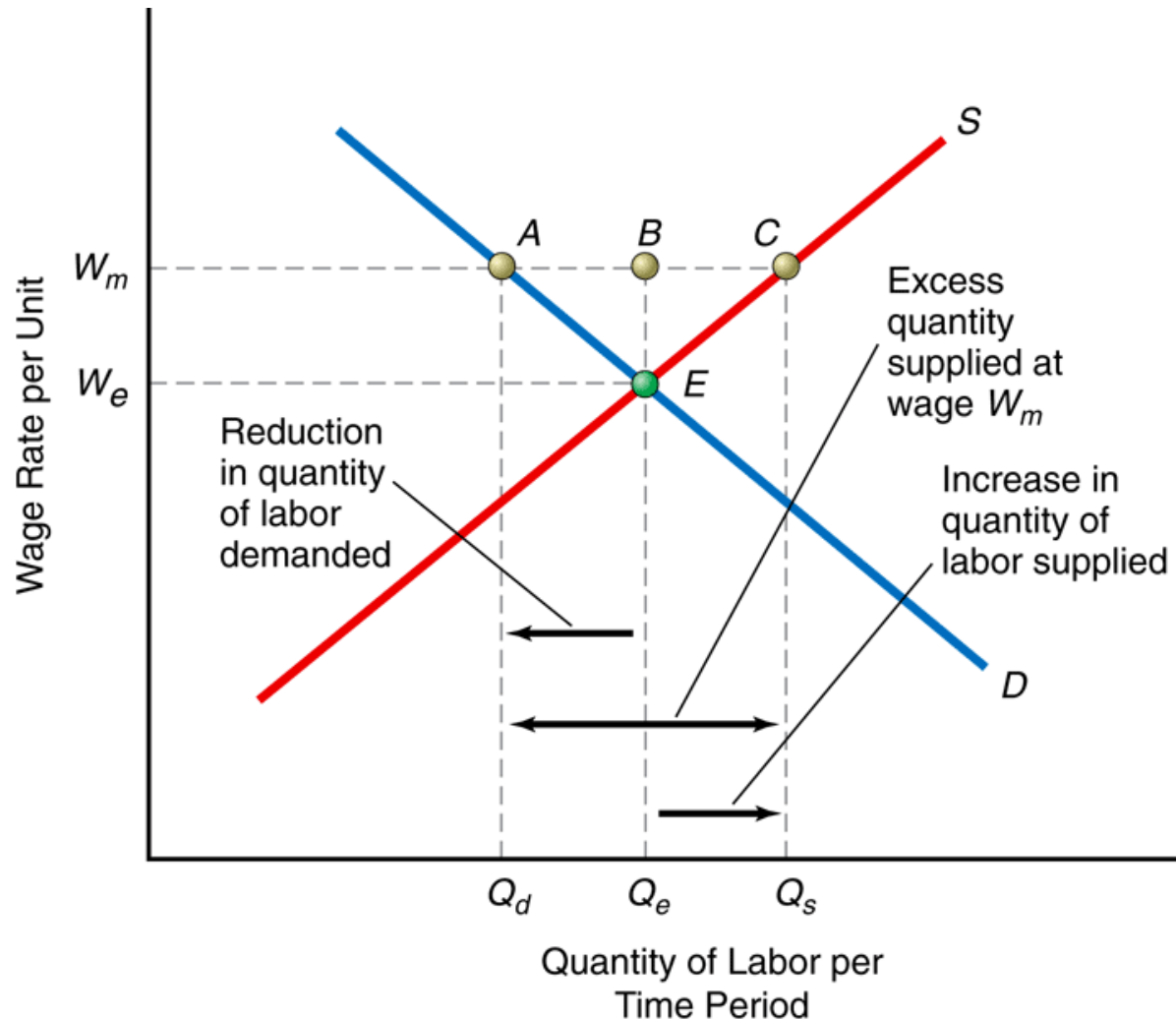
- How could the government keep the price from falling?
- Who benefits from agricultural price supports?

Price Floors in the Labor Market

Minimum Wage

- A wage floor, legislated by government, setting the lowest hourly wage rate that firms may legally pay their workers

The Effect of Minimum Wages



Quantity Restrictions

Governments can impose quantity restrictions, most obvious—banning ownership or trading of a good

- Human organs
- Drugs
- Hospital beds
- Gold from 1933 to 1973

Quantity Restrictions (cont'd)

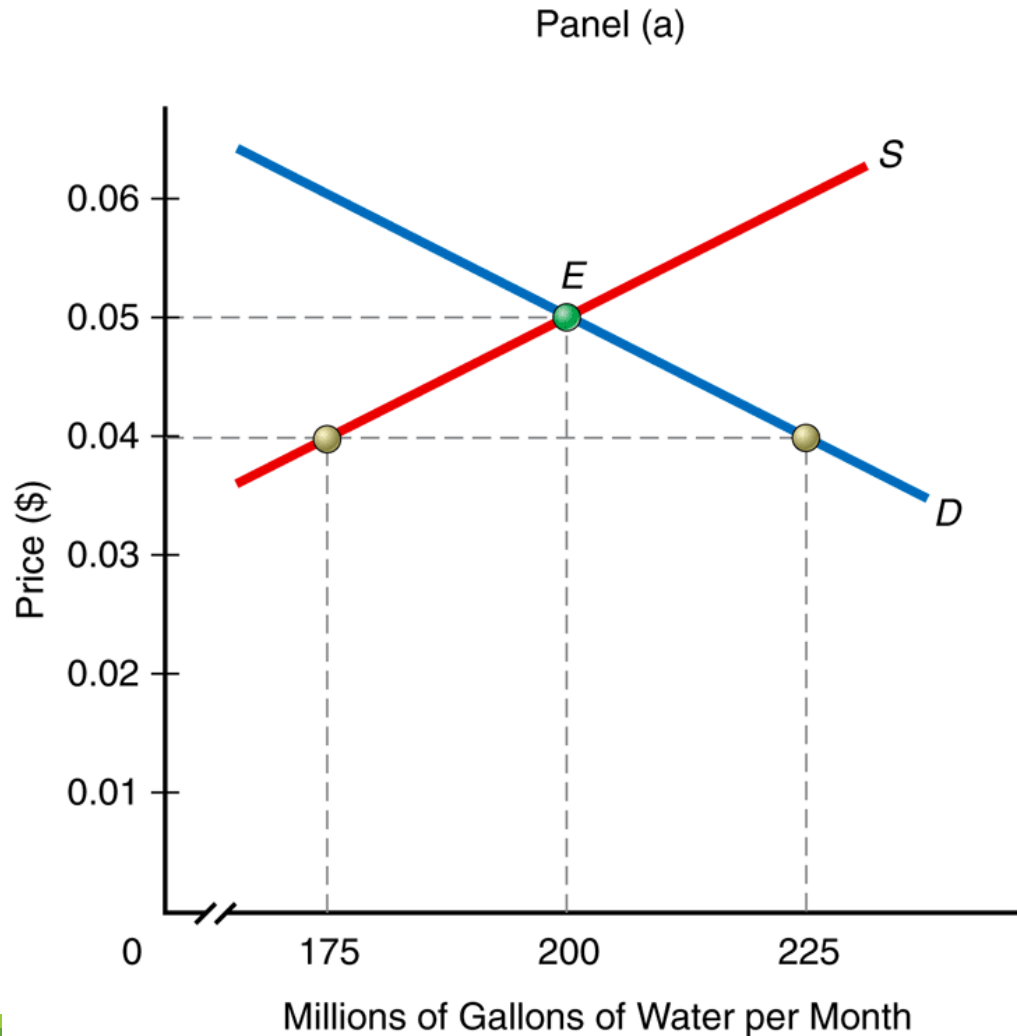
Government Prohibitions and Licensing Requirements

- Some commodities cannot be purchased at all legally; others require a license

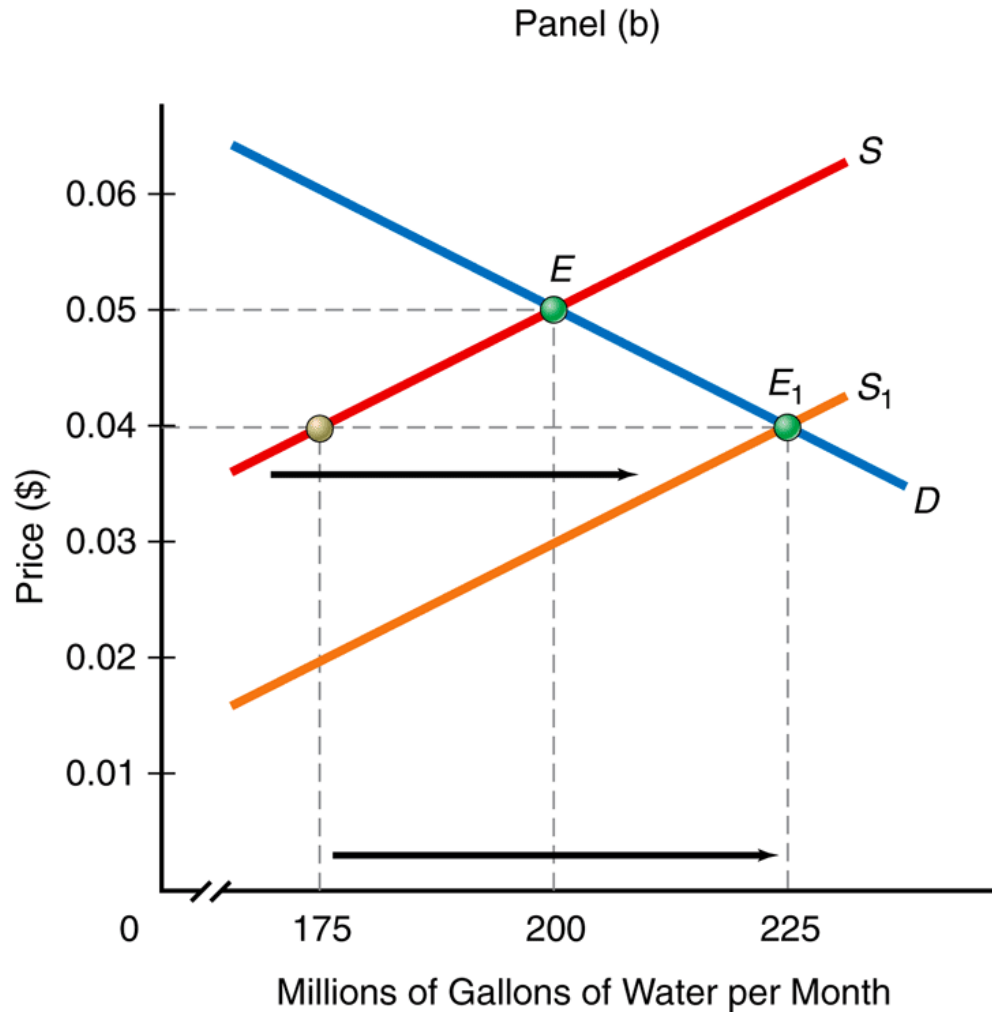
Import Quota

- Supply restriction that prohibits the importation of more than a specified quantity of a particular good

Two Methods of Eliminating a Water Shortage, panel (a)



Two Methods of Eliminating a Water Shortage, panel (b)



Summary

How changes in demand and supply affect market price and equilibrium quantity

- Increases in demand increase equilibrium price and quantity; decreases in demand decrease equilibrium price and quantity
- Increases in supply decrease market price and increase equilibrium quantity; decreases in supply increase market price and decrease equilibrium quantity

Summary

The rationing function of prices

- In a market system, prices ration scarce goods and services
- Other ways of rationing include first come, first served; political power; physical force; random assignment; and coupons

Summary

The effects of price floors

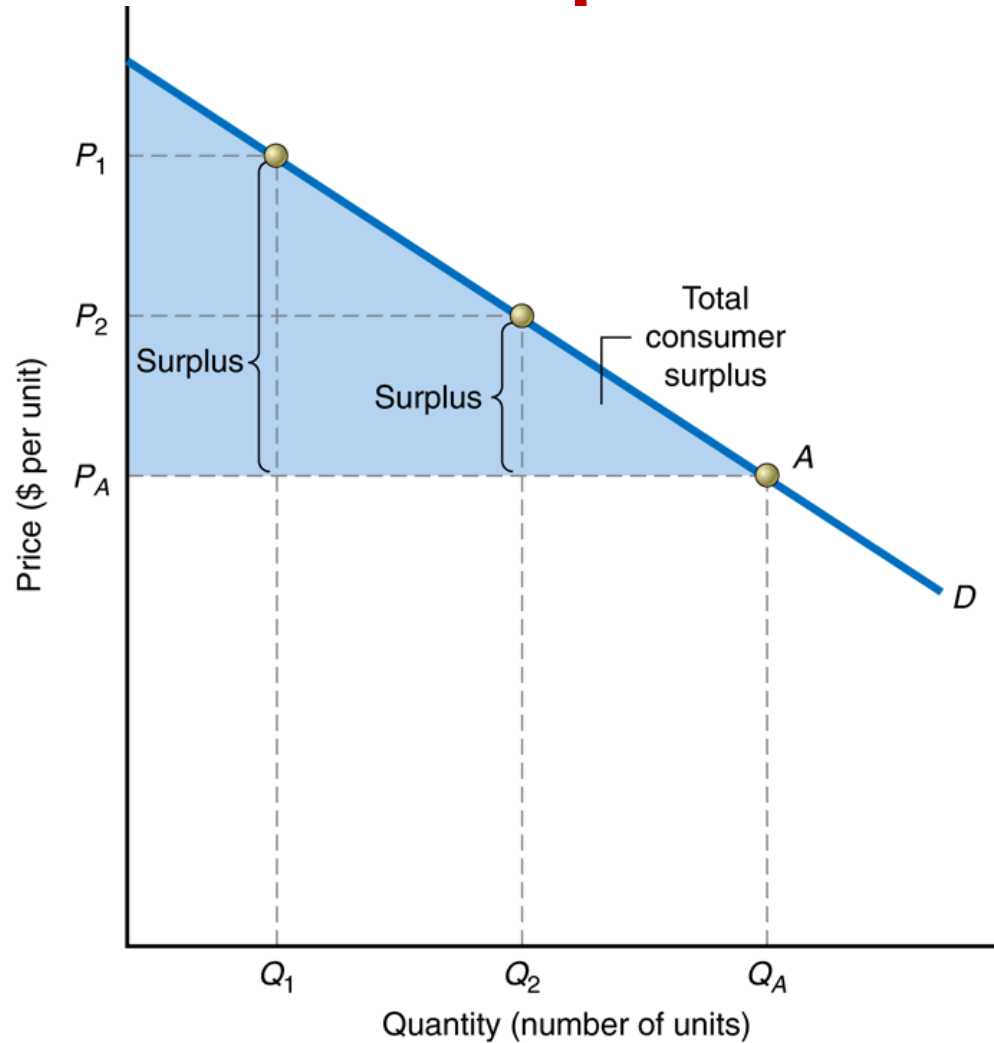
- If the price floor is set above the market clearing price, a surplus results
 - A price floor can take the form of a government-imposed price support or minimum wage

Consumer Surplus

Consumer Surplus

- The difference between the total amount that consumers would have been willing to pay for an item and the total amount that they actually pay

Consumer Surplus

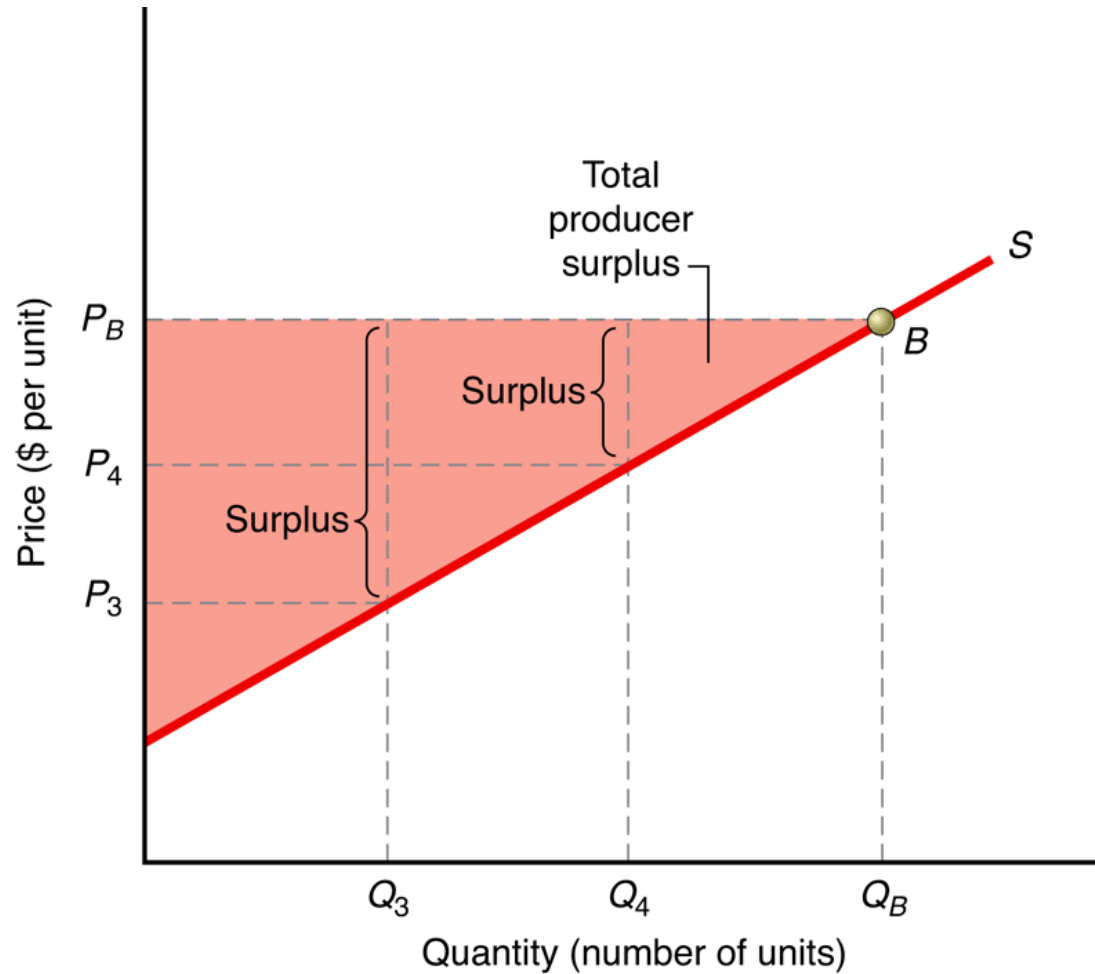


Producer Surplus

Producer Surplus

- The difference between the total amount that producers actually receive for an item and the total amount that they would have been willing to accept for supplying that item

Producer Surplus

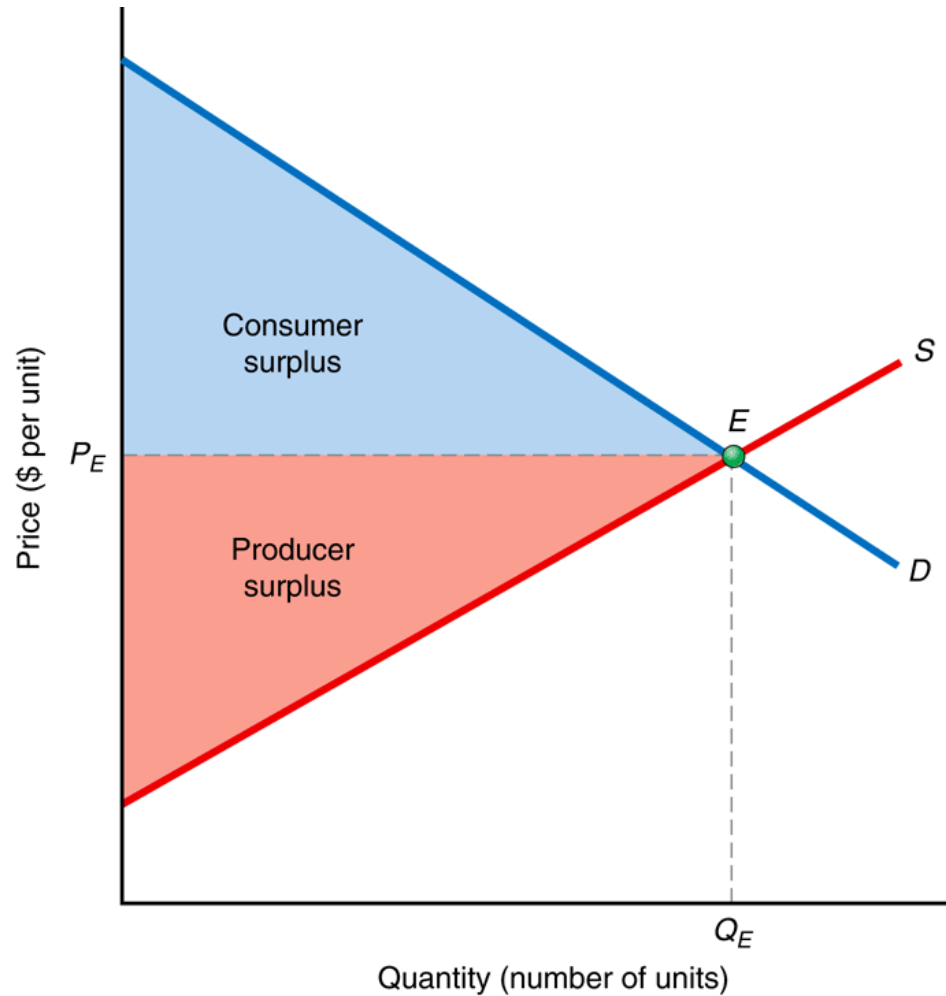


Gains from Trade within a Price System

Gains from trade

- The sum of consumer surplus and producer surplus

Consumer Surplus, Producer Surplus, and Gains from Trade



Price Controls and Gains from Trade

How do price controls affect gains from trade?

- Consumer surplus and producer surplus are both lower
- Either a price ceiling or a price floor reduces gains from trade